

Pengping Tan

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EDUCATION

Sun Yat-sen University

M.Eng. in Computer Technology
School of Computer Science and Engineering

Guangzhou, China
Sept. 2024 – Jun. 2027 (expected)

Sun Yat-sen University

B.S. in Information and Computing Science
GPA: 3.8/4.0 | Minor in Psychology

Zhuhai, China
Sept. 2020 – Jun. 2024

RESEARCH INTERESTS

Human-Computer Interaction, Human-Centered AI, Natural Language Processing

RESEARCH PROJECTS

JobMate: AI-Assisted Career Exploration with Real Stories from Social Media

Project Lead | Advisor: Prof. Zhenhui Peng

Guangzhou, China
Oct. 2025 – Apr. 2026

- Initiated and designed JobMate, an interactive system that transforms social media career stories into persona-grounded conversational agents to support career exploration, enabling users to reflect on and make sense of their own situations rather than defaulting to social comparison.
- Led the project end to end: conducted formative research comprising a 64-participant survey and eight semi-structured interviews, designed the data-processing pipeline and system interactions, and independently developed the full-stack system.
- Designed and conducted a between-subjects mixed-methods study with 24 participants and led the statistical and thematic analyses and manuscript preparation. Compared with native RedNote browsing, JobMate produced comparable reductions in career-decision difficulties while requiring significantly less effort on the NASA-TLX ($M = 3.92$ vs. 5.08 , $p = .012$). Qualitative findings showed that dialogue grounded in authentic experiences shifted users from upward social comparison toward self-reframing and sensemaking.

Investor Sentiment Analysis and Stock Price Prediction with Integrated Emojis

Team Lead and Co-First Author | Faculty Mentor: Prof. Patrick Houlihan, Columbia University

Remote
Jul. 2025 – Sept. 2025

- Led a four-person research team investigating whether emoji features could improve sentiment-aware stock-price prediction; developed the research plan, delegated tasks, and delivered biweekly progress presentations.
- Co-designed an LSTM-based multimodal model integrating FinBERT text embeddings, Emoji2Vec representations, and historical market indicators to predict next-day closing prices for three stocks using 10,000 StockTwits posts from the StockEmotions dataset.
- Led the literature review and manuscript preparation. Adding emoji features reduced MSE across all three stocks, with a 42.3% reduction for Boeing, resulting in a paper published in the ICFTBA 2025 proceedings.

Bai Juyi's Poetry and *Baishi Liutie*: A Computational Study

Undergraduate Thesis | Advisors: Prof. Baoquan Zhao & Prof. Zhitian Wang

Zhuhai, China
Nov. 2023 – May 2024

- Proposed and independently implemented a computational extension of Prof. Wang's manual study to identify textual connections between Bai Juyi's poetry and *Baishi Liutie*.
- Built a Python pipeline using rule-based text processing and exact matching to analyze 40 poetry volumes and 20,667 terms and allusions, identifying 1,246 matches and broadly reproducing the manual study's distributional trends.

PUBLICATIONS

- Tan, P., Zhao, B., & Peng, Z. (2026). Same Stories, Different Journeys: From Social Comparison to Sensemaking in AI-Mediated Peer Career Exploration. arXiv preprint; manuscript under revision. <https://arxiv.org/abs/2607.11039>
- Tan, P.^{*}, Chen, J.^{*}, Wu, R.^{*}, & Liu, H. (2025). Investor Sentiment Analysis and Stock Price Prediction with Integrated Emojis. ICFTBA 2025. ^{*}Equal contribution. <https://doi.org/10.54254/2754-1169/2025.GL28959>

TEACHING EXPERIENCE

Private Mathematics Tutor

High School Mathematics | One-on-One Instruction

China
Jan. 2024 – Jul. 2024

- Tutored nine high school students in weekly two-hour mathematics sessions, preparing individualized problem-solving lessons and adapting explanations to each student's needs and progress in both in-person and online settings.

AWARDS & SCHOLARSHIPS

- Outstanding Student Scholarship, Sun Yat-sen University: Second Prize (2023 – 2024); Third Prize (2021 – 2023)

SKILLS

- Research Methods:** Surveys, semi-structured interviews, mixed-methods user studies, statistical and thematic analysis
- Technical Skills:** Python, PyTorch, JavaScript, LLM APIs, Vue.js, Node.js, MongoDB, Neo4j
- Languages:** English (IELTS Academic: 7.0), Mandarin Chinese (native)